

What is this generically-named bacterial "membrane protein"? — Final Report

Summary

The protein annotated in UniProt as nothing more than a generic "**Membrane protein**" — **A0A0D5NHT9** from *Paenibacillus beijingensis* (314 aa), the representative of NCBI Protein Family **NF041162** — is **not a membrane protein at all**. It is a **Family 2A encapsulin nanocompartment shell protein** that adopts the **HK97 bacteriophage major-capsid-protein fold**. The "membrane protein" label is a **mis-annotation** that originates solely from an unreviewed, automated EMBL submission (accession **AJY74493.1**) and has no supporting structural or biochemical basis.

Three independent, converging lines of evidence establish this identity. First, the curated database signatures cross-referenced to this UniProt entry are unambiguous: NCBIfam **NF041162** carries the short-name "**encap_f2a**"; InterPro **IPR049822** is literally titled "**Encapsulin nanocompartment shell protein 2A**"; InterPro **IPR045641** is "Type 2A encapsulin shell protein SrpI-like"; and Pfam **PF19307** is the SrpI-like family. InterPro's own description states these proteins "share the HK97 fold of the capsid proteins" of *Caudovirales* bacteriophages. Second, the AlphaFold model (AF-A0A0D5NHT9-F1) is a high-confidence (global pLDDT 89.9) elongated $\alpha+\beta$ single domain with a large β -sheet and a long ~22-residue "spine" helix and — critically — **no transmembrane helix bundle**, exactly matching the HK97 capsid-subunit architecture and ruling out an integral membrane protein. Third, independent hydropathy analysis of the raw sequence shows a hydrophilic, negatively charged, soluble composition (net charge -6 ; maximum Kyte–Doolittle hydropathy of only 1.89, well below the ~2.5 threshold for a genuine transmembrane segment).

The correct higher-order assembly is a **self-assembling icosahedral protein shell** of ~60 or more identical subunits (a ~25–42 nm nanocompartment) that internally sequesters enzyme cargo. The correct subcellular localization / cellular component is the **Gene Ontology term GO:0140737 "encapsulin nanocompartment"** — an **intracellular, non-membrane-bound proteinaceous organelle** residing in the **cytoplasm**, explicitly *not* the plasma/inner

membrane. The single most decisive confirmatory experiment is **negative-stain or cryo-EM of the recombinantly expressed protein**, which should reveal ~25–42 nm icosahedral shells (with co-purified cargo density inside). Confidence in this conclusion is **very high**.

Key Findings

Finding 1 — A0A0D5NHT9 / NF041162 is a Family 2A encapsulin shell protein (HK97 fold), not a membrane protein

The decisive evidence is the set of curated, expert-reviewed database signatures cross-referenced to UniProt A0A0D5NHT9. These are not automated guesses; they are hand-curated protein-family models whose scope, membership, and functional descriptions are defined by domain specialists:

Signature	Accession	Name / description
NCBIfam (PGAP)	NF041162	short-name " encap_f2a " — the <i>equivalog</i> family in the research question
InterPro	IPR049822	" Encapsulin nanocompartment shell protein 2A "
InterPro	IPR045641	"Type 2A encapsulin shell protein SrpI-like"
Pfam	PF19307	SrpI-like encapsulin shell family

InterPro's curated narrative for these signatures states plainly that they describe "**capsid-like encapsulin nanocompartment shell proteins (type 2A)**" that "**share the HK97 fold of the capsid proteins**" found in *Caudovirales* bacteriophages, and that the founding member SrpI "**forms a nanocompartment probably involved in sulfur metabolism**." Every one of the four independent signatures points to the same conclusion — this is an encapsulin Family 2A shell subunit. This is precisely the Foldseek/structural conclusion the question anticipated: the fold **is** the self-assembling HK97 phage-capsid fold, reached directly through the protein's curated fold-family membership rather than through any membrane topology.

Independent sequence-based analysis fully corroborates a soluble shell protein and directly contradicts an integral-membrane assignment. A 19-residue sliding-window Kyte–Doolittle hydropathy scan gives a **maximum mean hydropathy of only 1.89** — no window approaches the ~2.5 threshold diagnostic of a bona fide transmembrane helix, and only two marginal

windows even exceed 1.6. The composition is that of a hydrophilic, charged protein: **net charge −6**, with **33 basic and 39 acidic residues** among 314 total. This is the amino-acid signature of a cytoplasmic, water-soluble protein that oligomerizes into a shell, not of a membrane-embedded polypeptide.

Crucially, the origin of the erroneous name is traceable: the UniProt display name "Membrane protein" derives **solely** from the unreviewed, automatically annotated EMBL nucleotide-derived protein entry **AJY74493.1**. It carries no experimental support, and no curated database signature endorses it. It is a placeholder that the specific NF041162/IPR049822/PF19307 assignments supersede.

The literature confirms that this is the correct protein class. Encapsulins are "**self-assembling protein nanocompartments widely distributed across prokaryotes that sequester diverse enzymes**" (PMID: 41481934); they "**self-assemble into icosahedral structures that capture cargo proteins through a cargo-loading peptide (CLP)**" (PMID: 42297566); and each is a "**protein nanocage consisting of a single monomer protein**" (PMID: 41794648) — i.e., the entire shell is built from one repeated subunit, so the "assembly" is an emergent property of self-association, not a membrane insertion.

Finding 2 — The AlphaFold model is a high-confidence $\alpha+\beta$ HK97-type capsid subunit, not a transmembrane helix bundle

The AlphaFold DB model **AF-A0A0D5NHT9-F1 (v6)** provides direct structural evidence, independent of the sequence-database signatures. Its global **pLDDT is 89.94**, with **76.4% of residues at very-high confidence** and only ~1% at very-low confidence — this is a **confidently folded single domain**, not a disordered or membrane-ambiguous chain for which AlphaFold would produce low confidence.

The geometry is diagnostic of a capsid subunit. The model is a **single elongated globular domain** with a radius of gyration of **28.0 Å** and a bounding box of **57.5 × 113.5 × 74.0 Å**. The pronounced long axis (~113 Å) is the hallmark of an HK97-type capsid subunit, which carries an extended **E-loop** and **P-arm/spine** that project along the shell surface.

The secondary-structure content, computed from backbone ϕ/ψ angles, is **mixed α/β dominated by an extensive β -sheet** (>20 β -strands) plus **6 helices, including one long ~22-residue helix** — the canonical HK97 "**spine helix**." Most importantly, there is **no bundle of long hydrophobic transmembrane helices**. An integral inner-membrane protein of 314 aa would present multiple contiguous ~20-residue hydrophobic helices spanning the model; their

complete absence, combined with the large β -sheet and single spine helix, is fully consistent with the HK97 bacteriophage-capsid fold assigned by IPR049822/PF19307 and is incompatible with a membrane protein.

Finding 3 — The correct cellular component is GO:0140737 "encapsulin nanocompartment," a non-membrane-bound organelle

Having established the fold and family, the correct subcellular localization follows directly and is captured by a dedicated Gene Ontology cellular-component term. **GO:0140737 "encapsulin nanocompartment"** (aspect: *cellular_component*) is defined as an **"Intracellular non-membrane bound organelle, consisting of proteinaceous polyhedral shells that encapsulate enzymes...** The self-assembling, 25–42 nm nanocompartment shell... is made of only one protein... The shell protein has an **HK97-like fold and probably evolved from a viral protein.**"

Because A0A0D5NHT9 is assigned to encapsulin Family 2A by four converging signatures (NF041162 "encap_f2a"; PF19307 SrpI-like; IPR049822 and IPR045641), its subunits localize to precisely this compartment: an **intracellular, non-membrane-bound, proteinaceous organelle in the cytoplasm**. This is the antithesis of the UniProt placeholder "Membrane protein." The organelle has no lipid bilayer, is not embedded in the plasma/inner membrane, and is not a membrane-spanning polypeptide. The GO framework thus provides the correct, specific cellular-component assignment and formally documents that "membrane protein" is a mis-annotation.

Mechanistic Model / Interpretation

Synthesizing the three findings yields a single coherent identity for A0A0D5NHT9 and the entire NF041162 family:

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Sequence: 314 aa, hydrophilic (net charge -6), no TM segment (max KD hydropathy 1.89)
    ↓
Fold: HK97 bacteriophage major-capsid-protein fold
      (α+β; large β-sheet + long ~22-aa "spine" helix; extended E-loop/P-arm)
      AlphaFold pLDDT 89.9, elongated subunit (Rg 28 Å, long axis ~113 Å)
    ↓
Family: Encapsulin Family 2A (NF041162 "encap_f2a";
      IPR049822 / IPR045641 SrpI-like; Pfam PF19307)
    ↓
Higher-order assembly: SELF-ASSEMBLING ICOSAHERAL SHELL
      ~60+ identical subunits → 25-42 nm nanocompartment
      interior encapsulates enzyme cargo via a cargo-loading peptide
    ↓
Localization: GO:0140737 "encapsulin nanocompartment"
      intracellular, NON-membrane-bound proteinaceous organelle (cytoplasm)
    ↓
Verdict: "Membrane protein" = MIS-ANNOTATION (from unreviewed EMBL AJY74493.1)

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The narrative is as follows. A single ~314-aa soluble polypeptide folds into an HK97-type capsid subunit — the same fold used by tailed bacteriophages (*Caudovirales*) to build their capsids, from which encapsulins are thought to have evolved. Many copies of this subunit self-assemble into a closed icosahedral shell 25–42 nm in diameter. This shell is a bacterial organelle — a **nanocompartment** — whose interior lumen sequesters specific enzyme cargo that is targeted to the shell through a cargo-loading peptide. For Family 2A / SrpI-like encapsulins, the associated chemistry is implicated in **sulfur metabolism**, though cargo identity varies across the family. The assembly floats in the cytoplasm as a non-membrane-bound organelle; it neither spans nor is anchored in the inner membrane.

The mis-annotation is therefore an artifact of automated pipelines that assigned a generic "Membrane protein" placeholder to an ORF (EMBL AJY74493.1) before any family-specific model matched it. Every subsequent, curated signal — NCBIfam, InterPro, Pfam, GO, AlphaFold structure, and raw hydropathy — overrides that placeholder in the same direction.

The following comparison table crystallizes why "membrane protein" fails and "encapsulin shell protein" succeeds:

Property	Expected for integral membrane protein	Observed for A0A0D5NHT9	Consistent with
Hydropathy	≥ 1 TM segment, KD > ~2.5	max KD 1.89; no TM window	Soluble shell
Net charge	Often near-neutral/hydrophobic core	−6 (hydrophilic, 33 K/R + 39 D/E)	Soluble shell
AlphaFold helices	Bundle of long hydrophobic TM helices	6 helices, one 22-aa spine; no TM bundle	HK97 capsid fold
AlphaFold topology	Membrane-spanning	Elongated $\alpha+\beta$ with large β -sheet	HK97 capsid fold
Curated family	(none supports membrane)	NF041162 / IPR049822 / PF19307 (encapsulin 2A)	Encapsulin shell
GO cellular component	plasma/inner membrane	GO:0140737 nanocompartment (non-membrane)	Cytoplasmic organelle

Evidence Base

The curated database signatures (NF041162, IPR049822, IPR045641, PF19307), the AlphaFold model (AF-A0A0D5NHT9-F1), the GO term GO:0140737, and the raw-sequence hydropathy analysis are the primary evidence; the literature below independently confirms that the encapsulin class this protein belongs to has exactly the properties (self-assembly, icosahedral shell, single-monomer nanocage, cytoplasmic nanocompartment) claimed here.

PMID	Title	How it supports the finding
42297566	<i>Encapsulin-Protected Immunotherapeutic Complexes: Bacteria-Derived Nanoparticles for mRNA Delivery to Eukaryotic Cells</i>	Establishes the icosahedral higher-order assembly and cargo encapsulation: "Encapsulins are protein nanocompartments that self-assemble into icosahedral structures that capture cargo proteins through a cargo-loading peptide (CLP)."
41794648	<i>Reprogramming the flavo-encapsulin from Thermotoga maritima to a novel protein nanocage platform for potent photodynamic therapy</i>	Confirms the shell is built from a single self-assembling monomer: "Encapsulins are protein nanocages consisting of a single monomer protein" — the assembly is emergent, not a membrane protein.
41481934	<i>Encapsulins in Terpene Biosynthesis: Enzyme Nanoreactors in Bacterial Secondary Metabolism</i>	Confirms encapsulins are the correct class: "Encapsulins are self-assembling protein nanocompartments widely distributed across prokaryotes that sequester diverse enzymes."
41309228	<i>Encapsulin protein MAV2054 enhances Mycobacterium avium virulence...</i>	Provides a concrete precedent that a " family 2A encapsulin nanocompartment shell protein " is a real, named functional entity in bacteria — the exact family assigned to AOA0D5NHT9.
41369466	<i>Characterization of Iron Oxide Nanoparticles Inside the [encapsulin]</i>	Reinforces that encapsulins are "microbial protein nanocompartments that spatially organize and sequester specific biochemical processes," consistent with a cytoplasmic organelle, not a membrane protein.
41239465	<i>Generation of hTERT-immortalized human MSCs with optical and magnetic labels...</i>	Independently describes bacterial nanocompartments as "high-molecular-weight icosahedral capsid-like protein complexes" with a shell + cargo architecture, matching the assembly model here.

No paper in the reviewed set challenges the conclusion; all are consistent with the encapsulin-shell identity. The literature does not, by itself, prove that *this specific* protein is an encapsulin — that is established by the curated signatures and structure — but it validates every property attributed to the family.

Supported vs. Refuted Hypotheses

- **Supported:** HK97 phage-capsid fold; self-assembling icosahedral encapsulin (Family 2A / SrpI-like) nanocompartment; cytoplasmic, non-membrane-bound localization; "membrane protein" is a mis-annotation.
- **Refuted:** Integral membrane protein / monomeric membrane component — no transmembrane helices in the AlphaFold model, no TM segment in the hydropathy profile, hydrophilic and negatively charged composition, and the curated fold family is a capsid shell.

Limitations and Knowledge Gaps

1. **No direct experimental structure of A0A0D5NHT9.** The fold assignment rests on curated homology models (Pfam/InterPro/NCBIfam) and an AlphaFold prediction, not an experimental cryo-EM or crystal structure of this exact protein. AlphaFold predicts the monomer fold with high confidence but does not itself prove shell assembly.
2. **Foldseek was not run as a live structural search in this investigation.** The HK97-fold assignment is supported by the AlphaFold monomer topology (large β -sheet + spine helix, no TM bundle) and by curated signatures that explicitly cite the HK97 fold, rather than by a fresh Foldseek query returning a specific PDB hit (e.g., an experimental HK97 or *T. maritima* encapsulin structure). A direct Foldseek run would strengthen the "closest experimentally characterized structural homolog" claim by naming a specific PDB and TM-score.
3. **Cargo identity is inferred, not demonstrated.** Family 2A / SrpI-like encapsulins are associated with sulfur metabolism, but the specific cargo enzyme(s) encapsulated by the *P. beijingensis* protein and its genomic neighbors were not characterized here.
4. **Assembly symmetry and T-number not determined.** Whether this shell is T=1 (~24 nm), T=3 (~32 nm), or T=4 (~42 nm) — and thus the exact subunit count — is not established for A0A0D5NHT9 and would require EM reconstruction.
5. **Species generalization.** The conclusion is drawn for the NF041162 representative; while the family is an *equivalog* (functionally conserved), individual members could in principle diverge, though this is unlikely given the tight signature agreement.

Proposed Follow-up Experiments / Actions

1. **(Decisive) Negative-stain and cryo-EM of the recombinant protein.** Express A0A0D5NHT9 in *E. coli*, purify, and image. The prediction is **~25–42 nm icosahedral shells** with internal cargo density, plus a single high-molecular-weight assembling species by size-exclusion chromatography / native PAGE. This single experiment would definitively confirm the self-assembling nanocompartment identity and refute "membrane protein." Cryo-EM single-particle reconstruction would additionally fix the T-number and subunit count.
2. **Run an explicit Foldseek search** of AF-A0A0D5NHT9-F1 against the PDB/AlphaFold databases to name the closest experimentally characterized structural homolog (expected: an HK97 phage capsid protein and/or a *T. maritima*/Family 2 encapsulin shell) with quantitative TM-score, closing limitation #2.
3. **Solubility / localization fractionation.** Cell fractionation (cytoplasm vs. membrane) with Western blotting or proteomics should place the protein in the **soluble cytoplasmic fraction**, not the membrane fraction — a fast, low-cost falsification test of the mis-annotation.
4. **Cargo identification.** Analyze the genomic neighborhood of the NF041162 locus (around EMBL AJY74493.1) for a co-encoded enzyme bearing a cargo-loading peptide (likely sulfur-metabolism-related for the SrpI-like Family 2A), and confirm co-purification of cargo with the shell by mass spectrometry.
5. **Database correction.** Submit evidence to UniProt to update the protein name from the placeholder "Membrane protein" to "Encapsulin shell protein (Family 2A)" and to annotate the cellular component as GO:0140737 "encapsulin nanocompartment."

Conclusion (one clear statement with confidence)

A0A0D5NHT9 / NF041162 is a Family 2A encapsulin nanocompartment shell protein with the HK97 bacteriophage major-capsid-protein fold. It self-assembles into a **~25–42 nm icosahedral, non-membrane-bound proteinaceous organelle (GO:0140737 "encapsulin nanocompartment")** in the cytoplasm that encapsulates enzyme cargo. **"Membrane protein" is a mis-annotation.** Confidence: **very high** (four converging curated signatures +

high-confidence AlphaFold $\alpha+\beta$ capsid fold with no TM helices + hydrophilic, charged, TM-free sequence). The one confirmatory experiment: **cryo-EM / negative-stain EM of the recombinant protein revealing ~25–42 nm icosahedral shells with internal cargo density.**

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